

Why this audit matters GARDEN HEALTH

PFAS compounds (per- and polyfluoroalkyl substances) enter the compost stream through food service packaging — pizza boxes, "compostable" cups, deli paper, fast-food wrappers. The composter processes the organic material but **cannot break down PFAS**: they are thermally stable at 55–70°C. Finished compost carries PFAS into garden soil, where plants absorb it through roots. Leafy greens — lettuce, spinach, kale, chard — accumulate the highest concentrations.

Colorado **HB 23-1034** (2023) prohibits PFAS in food packaging and food service ware by 2025. This audit helps businesses get ahead of that requirement and keeps contamination out of the compost stream now.

YOUR ROLE AS A VOLUNTEER

You are offering **compliance assistance**, not conducting an inspection. Most businesses don't know their "eco-friendly compostable" packaging contains PFAS. Come as a helper.

The water drop test DIY SCREEN

A quick field screen before visiting — or during, with permission:

- 1 Place **3 drops of water** on the inner surface of the packaging.
- 2 **Watch for 30 seconds.**
- 3 If water **beads and rolls** → likely fluorinated coating → flag for follow-up.
- 4 If water **absorbs quickly** → more likely PFAS-free → note as probable OK.

LIMITATIONS

The water test is a screen, not a lab result. Some PFAS-free biopolymer coatings also repel water. Suspected items should be confirmed with lab analysis (ASTM D8421). Record results in your audit form regardless.

High-priority items to audit

- **Pizza boxes**
Fluorinated inner surface is almost universal. Biggest volume contributor at most sites.
- **"Compostable" hot and cold cups**
BPI/OK Compost cert ≠ PFAS-free. Water barrier in paper wall often fluorinated.
- **Deli paper, sandwich wraps, butcher paper**
Grease barrier is key function; usually fluorinated. Test inner surface.
- **Paper plates (coated)**
Shiny/water-resistant surface. Uncoated kraft plates are usually fine.
- **Take-out containers (fibre/bagasse)**
Plain bagasse is usually PFAS-free. Oil-resistant versions may not be.
- **Unlined kraft paper bags**
Usually PFAS-free if no grease coating. Confirm with water test.

■ Red = high suspicion ■ Orange = moderate □ Green = likely OK

Business audit form RECORD & SUBMIT

Fill in during or immediately after visit. Submit to the SCD Hub bounty system or email to Cat at Town of Nederland.

Business name _____

Address _____

Date _____

Contact name _____

Business type _____

Packaging supplier _____

ITEM OBSERVED	WATER TEST	CURRENT BRAND	ACTION AGREED

HB 23-1034 awareness ☐ Not aware ☐ Aware, needs help ☐ Already switching

Notes

Conversation opener SCRIPT

OPENING (SLOW PERIOD — MID-MORNING OR MID-AFTERNOON)

"Hi — I'm a volunteer with the Nederland zero-waste programme, working alongside the town's sustainability manager, Cat. We're helping local businesses get ahead of Colorado's new food packaging rules. Most compostable packaging actually contains PFAS — a chemical that ends up in the compost and then in garden soil. We've got a list of PFAS-free alternatives. Do you have five minutes?"

IF THEY ASK "IS THIS A LEGAL VISIT?"

"No — this is purely voluntary. We're community volunteers trying to help businesses prepare before the 2025 deadline. Nothing we're collecting today goes to any regulatory body."

After the visit

- 1 **Email within one week** with specific PFAS-free alternatives matching their packaging needs (see reverse).
- 2 **Submit your audit form** to the SCD Hub platform to earn an eco:certificate credit for the visit.
- 3 **Confirmed switch?** Notify Cat — the business gets listed as a Clean Compost Partner.

PFAS-free packaging alternatives FOR BUSINESS

Share this list with businesses when you visit. Recommend they request SDS (Safety Data Sheet) or test documentation from their current distributor before ordering.

BRAND	CONFIDENCE	PRODUCT RANGE
World Centric	High — independently tested; results published 2021	Cups, plates, cutlery, take-out containers, bowls
Vegware	High — third-party tested; strong café/restaurant range	Hot cups, cold cups, plates, trays, clamshells
GreenPaper Products	High — PFAS-free certified line	Cups, plates; limited container range
Eco-Products	Good — verify specific SKUs; legacy products still in distribution	Broad range; request current PFAS-free line list
Unlined kraft bags	Always safe — no coating = no PFAS	Brown paper bags only; no grease resistance
Plain bagasse trays	Safe if uncoated — confirm no oil-resistance treatment	Sugarcane fibre plates, trays, clamshells
Reusable (stainless, glass)	Best — eliminates issue entirely	Any business considering a deposit-return programme

REUSE PROGRAMME OPPORTUNITY

Nederland's small size (approx. 1,500 residents, high tourist dwell time) makes it a strong candidate for a **community reuse container network** — deposit-and-return washable containers eliminating single-use packaging entirely. Boulder and Denver have piloted these. Cat's town sustainability role and vendor relationships are the exact institutional anchors such a programme needs. Contact SCD Hub to connect with running examples.

Colorado HB 23-1034 — summary for businesses LEGAL

YEAR	REQUIREMENT
2024	Manufacturers must <i>disclose</i> PFAS content to Colorado retailers and distributors. Businesses should be receiving updated SDS/disclosure forms from their packaging suppliers.
2025	Prohibition on PFAS in food packaging and food service ware. No new stock containing intentionally added PFAS may be sold in Colorado.
2026	Prohibition extends to additional consumer product categories including cosmetics, textiles, and cleaning products.

WHAT "INTENTIONALLY ADDED" MEANS

The prohibition applies to intentional use of PFAS in packaging. It does **not** guarantee zero PFAS from incidental contamination of raw materials. Testing is the only way to confirm a specific product is truly PFAS-free. Ask suppliers for third-party test documentation, not just a compliance declaration.

Community compost testing cycle FOR CAT / NEDERLAND

This is the technical backbone that volunteer advocacy feeds into. Results from each step are published transparently — on the waste transfer site noticeboard and on the SCD Hub platform.

- Baseline test (now):** Send finished compost from the EarthFlow system to Eurofins, ALS Environmental, or TestAmerica (all serve Colorado). Request **ASTM D8421** solid-matrix PFAS-40 panel. Cost: approx. \$200–350/sample.
- Input audit (months 1–3):** Volunteers document and photograph packaging in the organic waste stream using this form. Prioritise high-volume sources: restaurants, the school, any food trucks operating in season.
- Targeted substitution (months 3–9):** Work with businesses to switch highest-PFAS items. Focus on volume — one pizza restaurant contributing 10 boxes/night matters more than a boutique bakery contributing 3. Track confirmed switches on the platform.
- Re-test compost (month 12):** Same protocol as baseline. Compare total PFAS load. This validates the work quantitatively and earns eco:certificate credentials for participating volunteers.
- Publish and repeat:** Post results at the transfer site and email to participating businesses. Start the next 12-month cycle with a narrower target list.

Compost distribution gate UNTIL TESTING IS COMPLETE

PFAS LEVEL (TOTAL)	RECOMMENDED USE
< 2.5 ppb	Food gardens — label as tested and cleared
2.5–10 ppb	Avoid leafy greens; fruiting crops & root veg with caution; ornamentals OK
> 10 ppb	Hold for non-food uses: roads, revegetation, ornamentals, tree planting
Untested	Label "not yet cleared for food gardens — testing in progress"

SOIL MITIGATION IF PFAS FOUND

Activated carbon / biochar amendment (10–15% by volume) reduces plant-available PFAS by 30–60% in research settings. This is an interim tool for existing contamination — not a substitute for clean inputs. Increase soil organic matter to compete with PFAS at root absorption sites. Avoid growing leafy greens in affected plots until two consecutive clean tests.

Earn eco:certificate credits for this work

Completed business audit forms submitted to the SCD Hub platform (**exotopia.org**) qualify for eco:certificate credits — verifiable digital records of your contribution to community environmental health. Group coordinators can also post specific audit tasks as bounties: e.g., "Complete packaging audit at [business], document results, confirm follow-up action."

Contact: **acmeideal@gmail.com** · or through Cat at the Town of Nederland sustainability office

Bear Creek / South Boulder Creek watershed · Boulder County, Colorado